

VOLTAGE SENSING

# Initial Calculations

Pre-build design record — theoretical values to be corrected against real hardware measurements

|                |                                    |               |                                                     |
|----------------|------------------------------------|---------------|-----------------------------------------------------|
| Project        | Battery-Charge-Controller          | Branch        | step-03-voltage-sensing                             |
| Document       | docs/03-initial-calculations.md    | Date prepared | June 2026 (pre-build)                               |
| Battery config | 2S LFP (two 18650 cells in series) | Status        | <b>THEORETICAL — awaiting hardware verification</b> |

**Purpose:** Every value in this document is calculated from datasheets and theory before any hardware is assembled. When the breadboard is built and the multimeter calibration is done, a companion notes file (docs/03-calibration-notes.md) will record the real measured values. Discrepancies between these theoretical values and the measured values are expected and are part of the learning process.

## 1. System Voltage Range — What We Are Measuring

The battery pack used in this project is a 2S LFP (Lithium Iron Phosphate) configuration: two 18650 cells connected in series. The total pack voltage is the sum of the individual cell voltages at any given state of charge.

| Condition                       | Per Cell (LFP) | 2S Pack Total | Notes                                |
|---------------------------------|----------------|---------------|--------------------------------------|
| Fully discharged (cutoff)       | 2.50 V         | 5.00 V        | Do not discharge below this          |
| Nominal (resting, ~50% SoC)     | 3.20 V         | 6.40 V        | Typical operating midpoint           |
| Fully charged (CC-CV cutoff)    | 3.65 V         | 7.30 V        | CC-to-CV transition triggers here    |
| Absolute maximum (safety limit) | 3.70 V         | 7.40 V        | Overvoltage safety cutoff fires here |

Table 1: 2S LFP Pack Voltage Range

The ADC input on the Arduino Mega 2560 accepts 0–5.0 V only. The maximum pack voltage of 7.40 V exceeds this limit and would permanently damage the microcontroller if connected directly. A voltage divider is therefore required to scale the full voltage range down to within the ADC's safe input range.

## 2. Arduino Mega 2560 ADC Specifications

These are the ADC parameters that govern everything in this step. All calculations below derive from them.

| Parameter                    | Value                                                               | Source                  |
|------------------------------|---------------------------------------------------------------------|-------------------------|
| Resolution                   | 10 bits $\rightarrow 2^{10} = 1024$ steps (0 to 1023)               | ATmega2560 datasheet    |
| Reference voltage (default)  | AVCC = $\sim 5.0$ V (actual may vary: 4.85–5.10 V)                  | ATmega2560 datasheet    |
| LSB value (at exactly 5.0 V) | $5.0 / 1024 = 4.883$ mV per count                                   | Calculated              |
| Max safe input voltage       | 5.0 V (absolute maximum: 5.3 V)                                     | Arduino Mega datasheet  |
| Source impedance limit       | $\leq 10$ k $\Omega$ for full ADC accuracy                          | ATmega2560 datasheet    |
| Conversion formula           | $V_{\text{adc}} = (\text{rawValue} / 1024.0) \times V_{\text{ref}}$ | Note: 1024.0 not 1023.0 |
| Analogue pin used            | A0 (PA0)                                                            | Confirmed in firmware   |

Table 2: Arduino Mega 2560 ADC Parameters

**Why 1024.0 and not 1023.0:** The ADC produces 1024 equal steps. Step 0 = 0 V, step 1023 =  $V_{\text{ref}}$  minus one step (not  $V_{\text{ref}}$  itself). The formula  $(\text{raw}/1024.0) \times V_{\text{ref}}$  is therefore the correct mapping. Using 1023.0 introduces a +0.098% gain error — small but avoidable.

## 3. Voltage Divider Design — Step-by-Step Calculation

### 3.1 The Voltage Divider Formula

Two resistors R1 (upper, connected to battery +) and R2 (lower, connected to GND) form a divider. The voltage at their junction — the ADC input — is:

$$V_{\text{out}} = V_{\text{in}} \times (R2 / (R1 + R2))$$

The fraction  $R2 / (R1 + R2)$  is the divider ratio. It is fixed by the component values and is the same at every battery voltage. The relationship is perfectly linear — equal changes in  $V_{\text{in}}$  produce equal changes in  $V_{\text{out}}$  — making two-point calibration the correct and sufficient correction method.

### 3.2 Design Constraints

| Constraint                    | Value                         | Why                                            |
|-------------------------------|-------------------------------|------------------------------------------------|
| $V_{\text{in}}$ (max)         | 7.30 V (fully charged 2S LFP) | From Table 1                                   |
| $V_{\text{out}}$ (target max) | 4.50 V                        | 90% of 5.0 V — safety margin below ADC ceiling |

|                       |                                  |                                        |
|-----------------------|----------------------------------|----------------------------------------|
| V <sub>in</sub> (min) | 5.00 V (fully discharged 2S LFP) | From Table 1                           |
| Max source impedance  | ≤ 10 kΩ                          | ATmega2560 ADC accuracy requirement    |
| Resistor tolerance    | 1% (metal film)                  | 5% introduces ±146 mV systematic error |

Table 3: Design Constraints

### 3.3 Step A — Required Divider Ratio

$$\text{Ratio} = V_{\text{out}}(\text{max}) / V_{\text{in}}(\text{max}) = 4.50 / 7.30 = 0.6164$$

This means the ADC must always see 61.64% of the battery voltage.

### 3.4 Step B — Solve for R1 in Terms of R2

From the divider formula, rearranging for R1:

$$\begin{aligned} R1 / R2 &= (1 - 0.6164) / 0.6164 = 0.3836 / 0.6164 = 0.6224 \\ R1 &= 0.6224 \times R2 \end{aligned}$$

### 3.5 Step C — Choose R2, Solve for R1

Setting R2 = 10 kΩ (standard E24 value, also equal to the maximum allowable source impedance):

$$R1 = 0.6224 \times 10,000 = 6,224 \, \Omega$$

Nearest standard E24 resistor: 6.2 kΩ. Selected values: R1 = 6.2 kΩ, R2 = 10 kΩ.

### 3.6 Step D — Verify with Actual E24 Values

$$\begin{aligned} \text{Actual divider ratio} &= 10,000 / (6,200 + 10,000) = 10,000 / 16,200 = 0.6173 \\ V_{\text{out}} \text{ at } V_{\text{in}} = 7.30 \text{ V: } &7.30 \times 0.6173 = 4.506 \text{ V } \checkmark \text{ (below 5.0 V limit)} \\ V_{\text{out}} \text{ at } V_{\text{in}} = 5.00 \text{ V: } &5.00 \times 0.6173 = 3.087 \text{ V } \checkmark \text{ (ADC reading } \approx 631) \\ V_{\text{out}} \text{ at } V_{\text{in}} = 6.40 \text{ V: } &6.40 \times 0.6173 = 3.951 \text{ V } \checkmark \text{ (nominal midpoint)} \end{aligned}$$

### 3.7 Step E — ADC Resolution Over Measurement Range

$$\text{ADC span} = (4.506 - 3.087) / (5.0 / 1024) = 1.419 / 0.004883 = 290 \text{ counts}$$

$$\text{Voltage resolution} = (7.30 - 5.00) / 290 = 7.93 \text{ mV per ADC count}$$

The CC-to-CV transition must trigger at 7.30 V. A resolution of 7.93 mV per count means the transition is detected within one ADC step — approximately  $7.93 / 0.6173 = 12.85$  mV of battery voltage per count. This is more than adequate for reliable transition detection.

### 3.8 Step F — Power Dissipation Check

$$\begin{aligned} I_{\text{divider}} &= V_{\text{bat(max)}} / (R1 + R2) = 7.30 / 16,200 = 0.451 \text{ mA} \\ P_{R1} &= I^2 \times R1 = (0.000451)^2 \times 6,200 = 1.26 \text{ mW} \ll 250 \text{ mW rating} \\ P_{R2} &= I^2 \times R2 = (0.000451)^2 \times 10,000 = 2.03 \text{ mW} \ll 250 \text{ mW rating} \end{aligned}$$

Both resistors operate at less than 1% of their rated power. A standard 1/4 W resistor is used for both. No heatsinking required.

## 4. Theoretical Scaling Factor (Pre-Calibration)

The firmware converts the averaged ADC reading back to a battery voltage using this scaling factor. Before calibration, the theoretical value is derived from the resistor values:

$$\text{Scaling factor (theoretical)} = (R1 + R2) / R2 = 16,200 / 10,000 = 1.620$$

This is the SLOPE constant entered in the firmware before calibration. After calibration against the multimeter (Section 6), this value will be replaced by the measured correction.

$$\begin{aligned} V_{\text{battery (theoretical)}} &= V_{\text{adc}} \times 1.620 \\ \text{Full formula: } V_{\text{battery}} &= ( \text{rawADC} / 1024.0 \times V_{\text{ref}} ) \times 1.620 \end{aligned}$$

**Pre-calibration accuracy estimate:** With 1% tolerance resistors, the worst-case resistor error alone is  $\pm 1\%$  gain error =  $\pm 73$  mV at 7.30 V. The AVCC reference can vary  $\pm 2\%$ , adding another  $\pm 146$  mV. Combined worst case before calibration: approximately  $\pm 220$  mV. After two-point calibration against a multimeter, residual error should be below  $\pm 30$  mV.